Mikael Geraldius Gunawan

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EDUCATION

Universitas Multimedia Nusantara, Bachelor's in Informatics, GPA 3.94/4.00 *2024 - Present (expected 2028)*
SMA Pelita Kasih, Senior High School *2021 - 2024*

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, Dart, PHP, Kotlin, C#
Frontend: React, Next.js, Inertia.js, Vite, Tailwind CSS, HTML, CSS, HTML Canvas, Web Audio API, Figma
Mobile: Flutter, React Native, Expo, Kotlin with Room, background location, push notifications
Machine Learning & AI: pandas, scikit-learn, numpy, Random Forests, evaluation against baselines, Google Gemini API
Backend: Laravel, NestJS, Supabase (Row Level Security), PostgreSQL, MySQL, Express.js
Cloud & DevOps: Cloudflare Workers, Supabase Edge Functions, GitHub Actions (CI/CD), GitHub Pages, Linux
Tools: Git, Claude Code
Spoken Languages: Indonesian (native), English (fluent)

WEB DEVELOPMENT EXPERIENCE

UMN Festival **Tangerang, Indonesia**
Member of Website Division (Portal and Pixels) | React, Inertia.js, Laravel, Tailwind CSS *Jan 2026 - Present*

- Built the event cards, countdown and introduction section of the Event page on umnfestival.com, the site for UMN's yearly festival.
- Made the cards and countdowns read from the site's API so the committee can edit them in the admin panel.
- Worked in a nine-person repository with a branch per page, merging through pull requests and resolving conflicts.

UMN ECO **Tangerang, Indonesia**
Member of Website Division (Backend and Frontend) | NestJS, Next.js *Jan 2026 - Present*

- Built login and ticket management in NestJS for the organization's website, with a database that tracks each user's ticket count.
- Worked on the website's About page in Next.js.

Duta Anti Narkoba UMN **Tangerang, Indonesia**
Coordinator of Website Division | React, TypeScript, Vite, Tailwind CSS *Sep 2025 - Nov 2025*

- Led the website team from start to finish on a site with organization profiles, event schedules and ticketing.
- Ran the team's work through GitHub and helped members learn React and Tailwind CSS.

PROJECTS

Song Mood Classifier | Machine Learning final project, UMN **2026**
Python, pandas, scikit-learn, TypeScript, React, Vite, Cloudflare Workers *mikaelgeraldius.github.io/mood-classifier*

- Built the web version of my group's final project: a Random Forest that names a song's mood (sad, happy, energetic or calm) from nine audio features, running entirely in the browser.
- Reported the model's 92.3% accuracy on 55,588 unseen songs next to two baselines: logistic regression at 83.5% and always guessing "happy" at 38.3%.
- Cut the forest from 9.2 million nodes to 357 thousand (a 1.4 MB file) for about one point of accuracy, and built a TypeScript reader that agrees with scikit-learn on 99.95% of test songs.
- Switched to ReccoBeats behind a Cloudflare Worker when Spotify closed audio features to new apps, and added a fallback that measures loudness and tempo from the audio in the browser.
- Developed the web version with Claude Code and deployed it through a GitHub Actions workflow that runs the tests and builds the site on every push to main.

ReLa | JOINTS UGM 2026 hackathon, team of four **2026**
Flutter, Dart, Supabase Edge Functions, PostgreSQL, TypeScript, Google Gemini, Figma *github.com/AltoDevv/rela-submission*

- Brainstormed ideas with Claude, then designed the screens in Figma and coded an Android app that links older parents living alone with community volunteers through local foundations, so families far away can follow each visit.
- Volunteers describe a visit in plain words and Gemini sorts it into seven health areas, quoting the sentence behind each; on 35 test stories it got 88.6% of areas right and caught all 18 danger signs.
- Ran the Gemini calls in Supabase Edge Functions, checked the output against a schema, stripped names and addresses before sending stories, and kept each role's data apart with Row Level Security.

RegionMusic | Personal Android app **2026**

TypeScript, React Native, Expo, OpenStreetMap, Python, numpy, ffmpeg *github.com/MikaelGeraldius/RegionMusic*

- Built an Android app that follows your location in the background, even with the phone locked, and crossfades to a theme for the kind of place you are in: malls get a city theme, housing complexes a town theme, roads a route theme.
- Classified places from OpenStreetMap with point-in-polygon tests and smoothed GPS jitter so the music doesn't flicker at borders.
- Found each track's loop point in Python with autocorrelation (numpy, ffmpeg), so songs repeat with no fade or gap.
- Planned and built it in phases with Claude Code, and tested it on a real phone, fixing what broke there.

Wemen | Pixel-art browser game, UMN final project **2025**

JavaScript, React, Vite, HTML Canvas *mikaelgeraldius.github.io/UAS-WEMEN*

- Made a life-sim game with one teammate where you keep energy, hunger, hygiene and mood above zero for ten days in a town.
- Drew the town on a canvas every frame inside React, with tile-map collisions and all game rules kept in one reducer.
- In 2026, used Claude Code to rebuild the HUD for phones (on-screen arrows and swipe controls) and rewrite the snake mini game.

WORK EXPERIENCE

Excellent Organizer | Wedding organization **Jakarta Barat, Indonesia**

Multimedia, Entertainment and Sound Division *Jan 2025 - Present*

- Manage vendors and delegate tasks to coworkers for weddings.

LEADERSHIP & ORGANIZATION

HMIF (Himpunan Mahasiswa Informatika) UMN **Tangerang, Indonesia**

Coordinator of Project Manager Division *Dec 2025 - Present*

- Coordinated the events division for Inforta, the faculty's anniversary event, where students play games together.
- Supervised FUSE, a non-academic competition with Valorant, Mobile Legends, basketball and badminton tournaments.
- Organized the regular PLENO meetings where each HMIF division reports its progress.
- Organized company visits where students see how companies work and what systems they run on.

PPIF (Perkenalan Prodi Informatika) UMN **Tangerang, Indonesia**

Member of Module Division, PIC for new Informatics students *Feb 2025 - Sep 2025*

Antiokh | Church organization run by teenagers **Jakarta Barat, Indonesia**

Parish Coordinator (Korpar), previously Acara (Events) Division *Jul 2024 - Present*

- Guide new members as Korpar since Sep 2026. In Acara, made event rundowns, led teams, presented and spoke in public.